From Person-Specific Dynamics to Population Inference: A Computationally Efficient Meta-Analytic Structural Equation Modeling Framework for Integrating Complex Multilevel Dynamic Models

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Computations for this research were performed on the Pennsylvania State University's Institute for Computational and Data Sciences' Roar supercomputer using SLURM for job scheduling (Yoo et al., 2003), GNU Parallel to run the simulations in parallel (Tange, 2021), and Apptainer to ensure a reproducible software stack (Kurtzer et al., 2017, 2021).

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Links

- **Research compendium:** OSF, GitHub, and the compendium website.
- **R packages:** `fitVARMxID` for person-specific estimation and `metaDyn` for meta-analytic synthesis; source code is available at `fitVARMxID` on GitHub and `metaDyn` on GitHub.
- **Worked examples:** one simulation replication, including Mplus inputs, priors, and diagnostics; the empirical analysis; and the computational benchmark.
- **Reproduction:** container instructions.

Additional Information

Prior Sensitivity

The simulation compares dynamic structural equation modeling (DSEM) under default and alternative priors. Defaults use diffuse normal location priors and $\text{IW}(\mathbf{0}, -d-1)$ for covariance blocks of dimension d (Asparouhov & Muthén, 2023). Alternatives use $N(0, 25)$ for set-point means, $N(0, 0.25)$ for transition means, $\text{IW}(\mathbf{I}_2, 3)$ for set-point and innovation covariances, and $\text{IW}(0.004\mathbf{I}_4, 5)$ for transition covariance. Normal-prior second arguments are variances. The alternative transition-variance marginals have mode .001 and no finite mean; these proper, heavy-tailed priors assess sensitivity near small variances rather than eliminate prior influence. The simulation vignette provides example inputs and prior printouts.

Usable Replications and Boundary Estimates

Table 1 reports usable replications and boundary frequencies for every condition. MetaVAR estimates $\boldsymbol{\tau}^2 = \mathbf{LDL}'$ using unit lower-triangular $\mathbf{L}$ and softplus-transformed diagonal entries plus 10^{-8} in $\mathbf{D}$. Otherwise usable near-boundary fits are retained. Near-zero variances and near-singular covariance matrices are distinct from inadmissible fits; neither positive parameterization nor robust Wald intervals guarantees valid inference at zero. Figure 6 reports zero-target exclusion separately from nonzero detection. Bayesian variance intervals exclude zero by construction.

Empirical Comparison

MetaVAR and the uncertainty-uncorrected benchmark (Uncorr) use identical nine-feature Stage 1 estimates for the 212 retained participants. MetaVAR also uses the full joint sampling covariance; a positive-definite approximation was applied when its smallest eigenvalue was at or below 10^{-6} (43 participants; maximum entry change 2.40×10^{-5}). Both empirical DSEM fits terminated normally but had inadequate posterior convergence: rank-normalized $\widehat{R}$ for the positive-affect inertia variance was 1.442 under default priors and 1.462 under alternatives, with bulk effective sample sizes below eight. Inertia

means also failed the posterior-quality screen. These fits therefore cannot support a reliable numerical comparison; their saved output, full diagnostics, and trace plots remain in the compendium. The shared-input MetaVAR–Uncorr comparison in the manuscript illustrates the effect of uncertainty propagation, without establishing accuracy against unknown population values.

Simulation Performance Figures

Indices 1–6 denote $(\mu_1, \mu_2, a_{11}, a_{21}, a_{12}, a_{22})$; α denotes population means and τ^2 between-person variances/covariances. Figures include all 36 conditions and both DSEM priors. Unbalanced denotes cyclic lengths of 50, 100, and 200. Recovery uses calibrated targets after stability rejection. MetaVAR uses normal-theory intervals. Relative bias and detection apply to nonzero targets; zero targets use absolute bias and separately displayed interval exclusion. Grey tiles denote undefined or unavailable metrics.

References

Asparouhov, T., & Muthén, B. O. (2023). *Bayesian analysis using Mplus: Technical implementation*. Tech. rep. http://www.statmodel.com/download/Bayes2.pdf

Kurtzer, G. M., cclerget, Bauer, M. W., Kaneshiro, I., Trudgian, D., & Godlove, D. (2021). *hpcng/singularity: Singularity 3.7.3*. Zenodo. https://doi.org/10.5281/zenodo.4667718

Kurtzer, G. M., Sochat, V., & Bauer, M. W. (2017). Singularity: Scientific containers for mobility of compute. *PLOS ONE*, *12*(5), e0177459. https://doi.org/10.1371/journal.pone.0177459

Tange, O. (2021). *GNU Parallel 20210922 ('Vindelev')*. Zenodo. https://doi.org/10.5281/zenodo.5523272

Yoo, A. B., Jette, M. A., & Grondona, M. (2003). SLURM: Simple Linux Utility for Resource Management. In D. Feitelson, L. Rudolph, & U. Schwiegelshohn (Eds.), *Job scheduling strategies for parallel processing* (pp. 44–60). Springer Berlin Heidelberg. https://doi.org/10.1007/10968987_3

Table 1

Usable Replications, MetaVAR Boundaries, and DSEM Posterior Quality

Cell	h	N	T	Meta usable	Near singular	Near zero	DSEM-D pass	DSEM-A pass
1	1	50	50	1000	0.635	0.000	477	0
2	1	100	50	1000	0.323	0.000	95	0
3	1	200	50	1000	0.092	0.000	0	0
4	1	50	100	1000	0.629	0.000	610	7
5	1	100	100	1000	0.373	0.000	242	2
6	1	200	100	999	0.183	0.000	58	0
7	1	50	200	1000	0.474	0.000	869	508
8	1	100	200	1000	0.262	0.000	681	476
9	1	200	200	1000	0.132	0.000	424	348
10	1	50	U	1000	0.618	0.000	448	7
11	1	100	U	1000	0.352	0.000	218	10
12	1	200	U	1000	0.154	0.000	71	6
13	2	50	50	999	0.483	0.000	562	0
14	2	100	50	1000	0.195	0.000	173	0
15	2	200	50	1000	0.047	0.000	22	0
16	2	50	100	1000	0.445	0.000	716	42
17	2	100	100	1000	0.196	0.001	450	27
18	2	200	100	1000	0.073	0.000	231	7
19	2	50	200	1000	0.259	0.000	910	673
20	2	100	200	1000	0.077	0.000	834	589
21	2	200	200	1000	0.014	0.000	750	429
22	2	50	U	1000	0.453	0.000	546	48
23	2	100	U	1000	0.202	0.000	379	59
24	2	200	U	1000	0.042	0.000	266	23
25	0	50	50	996	0.980	0.006	138	0
26	0	100	50	997	0.936	0.003	2	0
27	0	200	50	998	0.797	0.001	0	0
28	0	50	100	991	0.998	0.015	205	0
29	0	100	100	992	0.989	0.007	6	0
30	0	200	100	995	0.959	0.002	0	0
31	0	50	200	983	0.996	0.020	413	0
32	0	100	200	990	0.995	0.026	41	0
33	0	200	200	982	0.993	0.013	0	1
34	0	50	U	982	0.996	0.015	42	0
35	0	100	U	992	0.989	0.016	0	0
36	0	200	U	986	0.982	0.009	0	0

Note. Each cell requests 1,000 replications. All Stage 1, Uncorr, and both DSEM branches have 1,000 recorded usable outputs; MetaVAR retains 35,882 of 36,000 (118 inadmissible). U denotes cyclic unequal lengths; h is the transition-covariance multiplier. DSEM-D and DSEM-A denote default and alternative priors. Near singular means minimum covariance eigenvalue below 10^{-8}; near zero means a diagonal variance at most 10^{-6}. These rates condition on retained fits and are distinct from failures. DSEM passing counts require every diagnostic to be finite, $\widehat{R} \leq 1.01$, bulk and tail effective sample sizes ≥ 400, and Monte Carlo error relative to posterior SD ≤ 0.05. They do not replace primary denominators.

Figure 1

Overall Performance

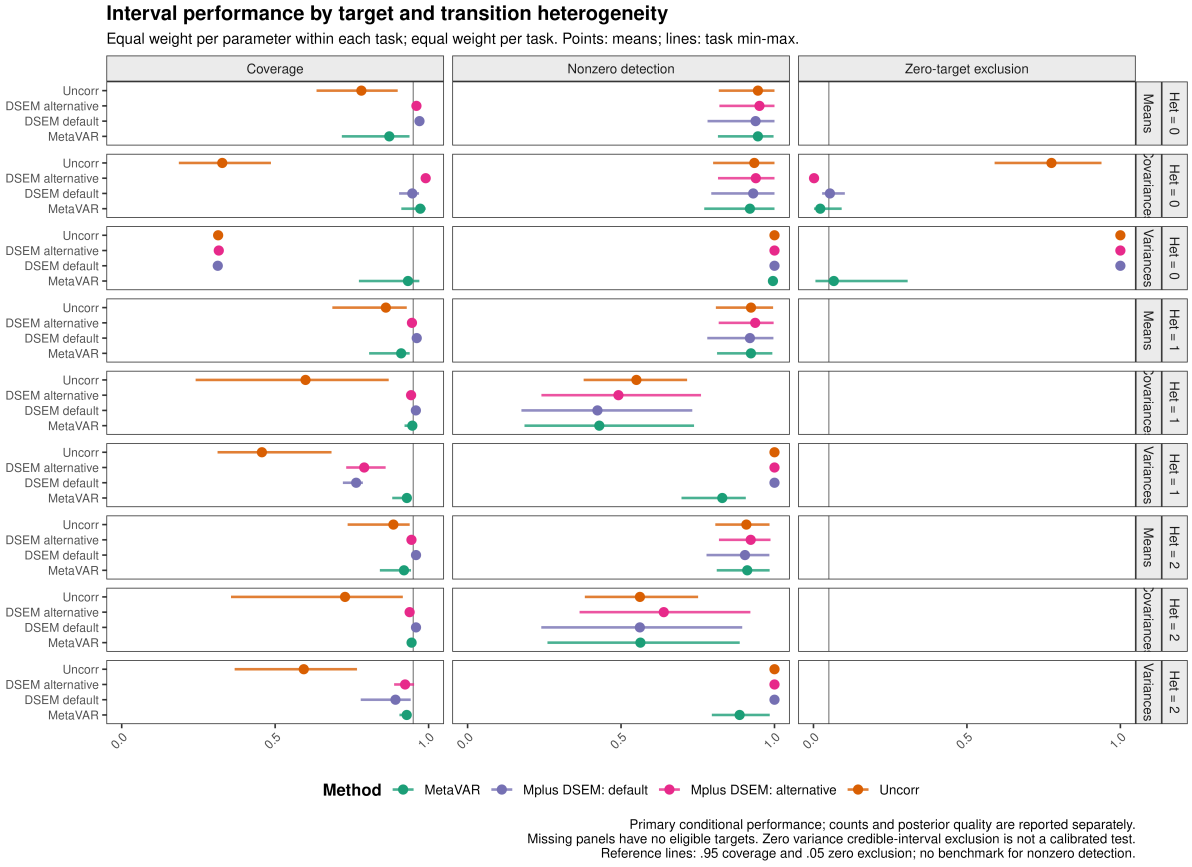


Figure 2

Coverage Across Transition-Heterogeneity Conditions

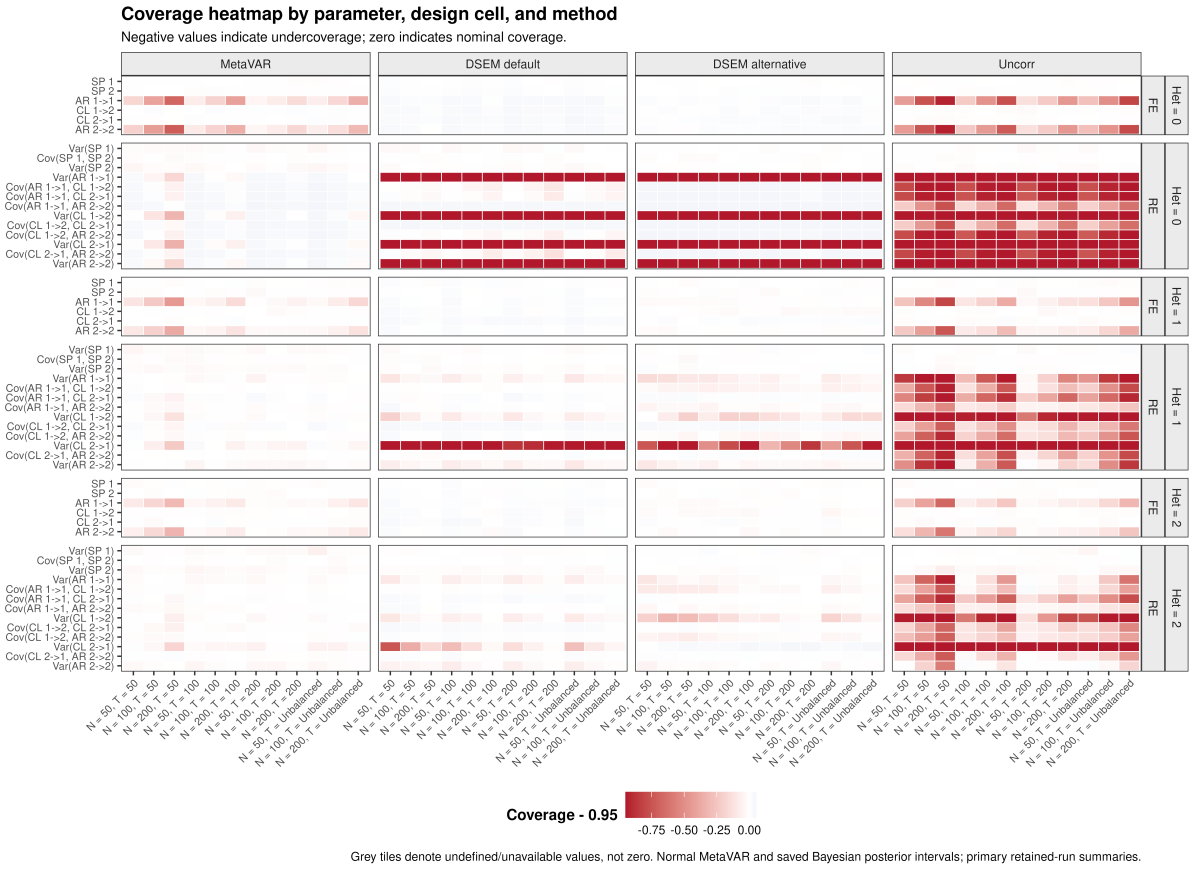
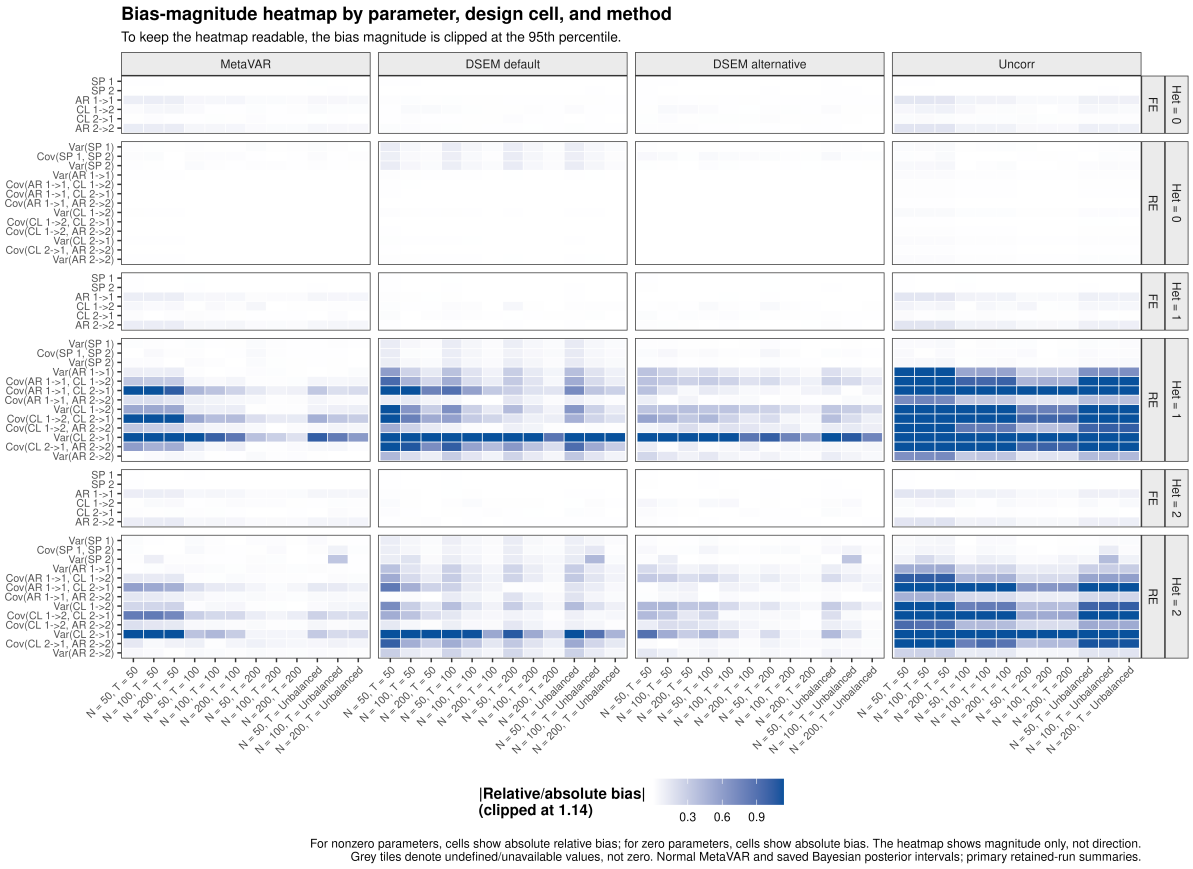


Figure 3

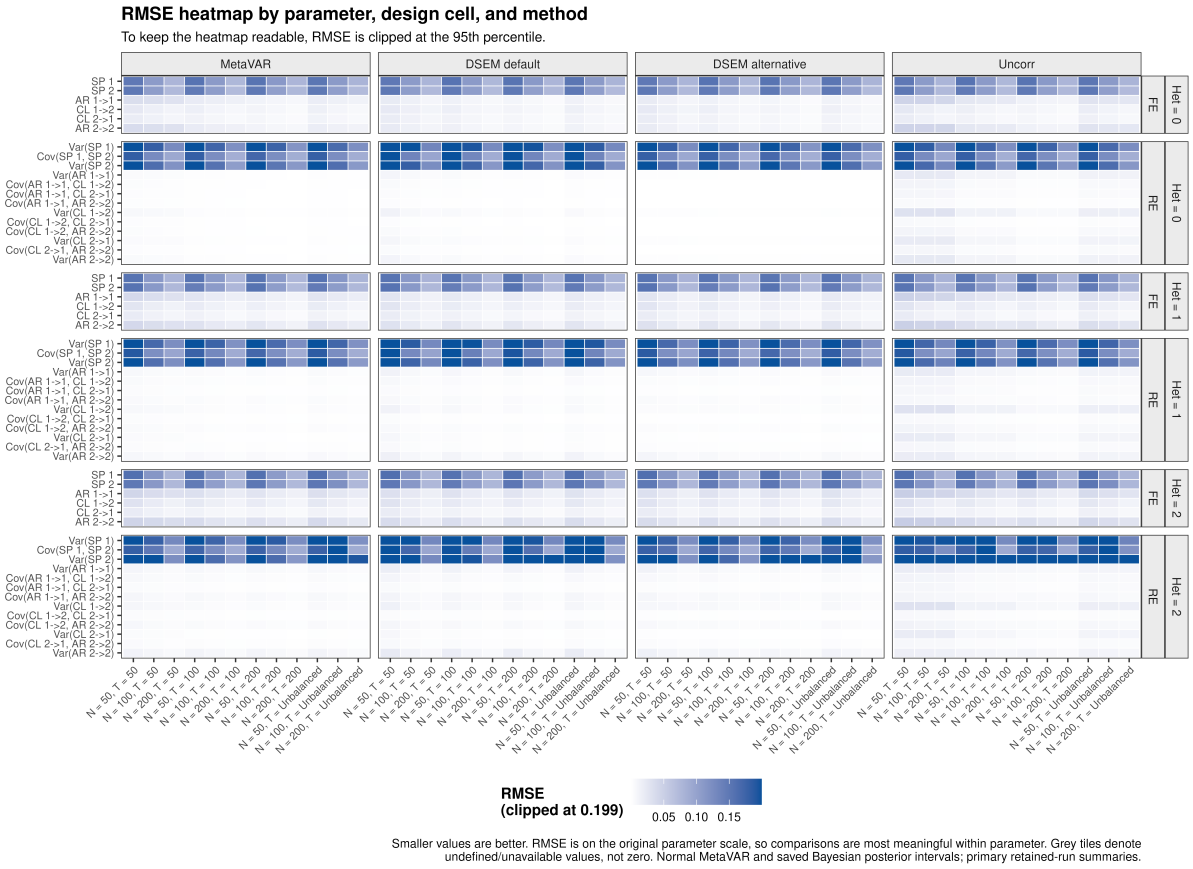
Absolute Relative or Absolute Bias Across Transition-Heterogeneity Conditions



For nonzero parameters, cells show absolute relative bias; for zero parameters, cells show absolute bias. The heatmap shows magnitude only, not direction. Grey tiles denote undefined/unavailable values, not zero. Normal MetaVAR and saved Bayesian posterior intervals; primary retained-run summaries.

Figure 4

Root Mean Square Error Across Transition-Heterogeneity Conditions



Smaller values are better. RMSE is on the original parameter scale, so comparisons are most meaningful within parameter. Grey tiles denote undefined/unavailable values, not zero. Normal MetaVAR and saved Bayesian posterior intervals; primary retained-run summaries.

Figure 5

Detection of Nonzero Targets Across Transition-Heterogeneity Conditions

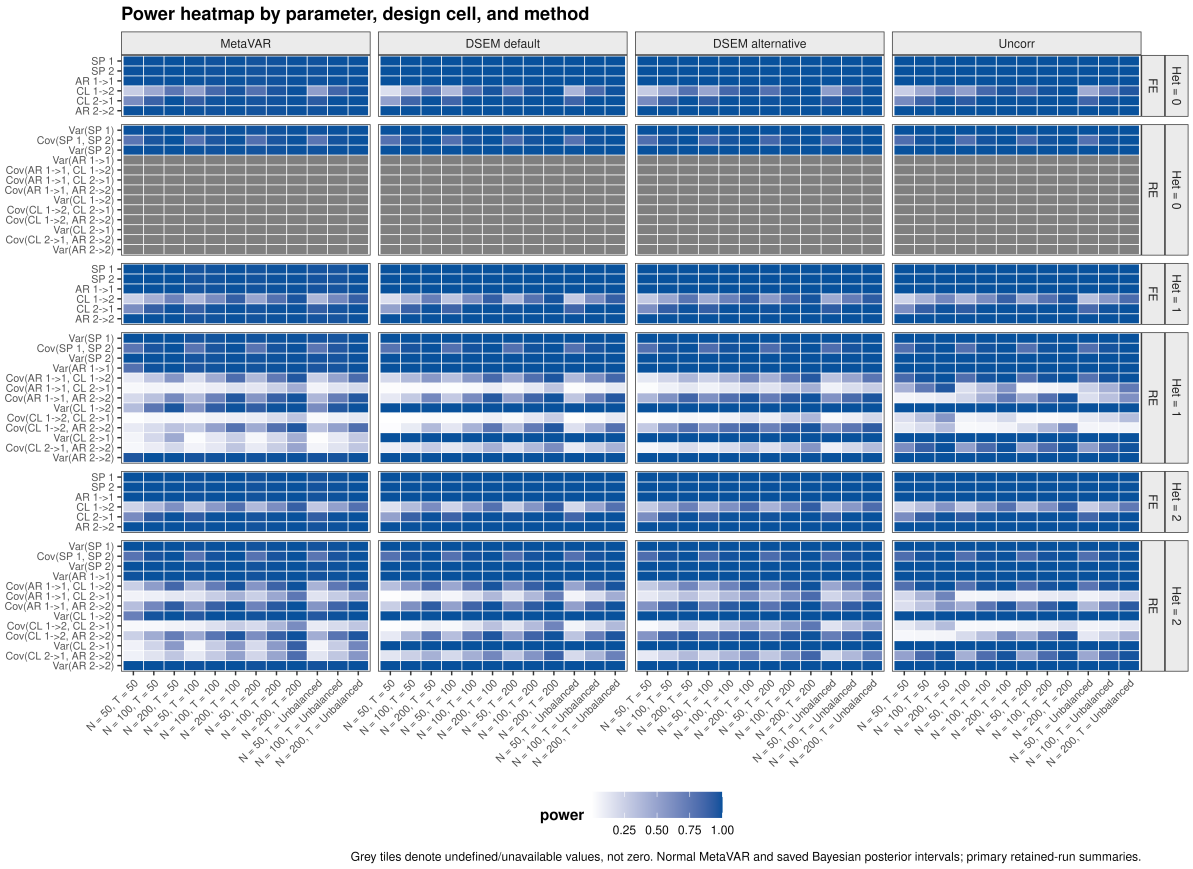
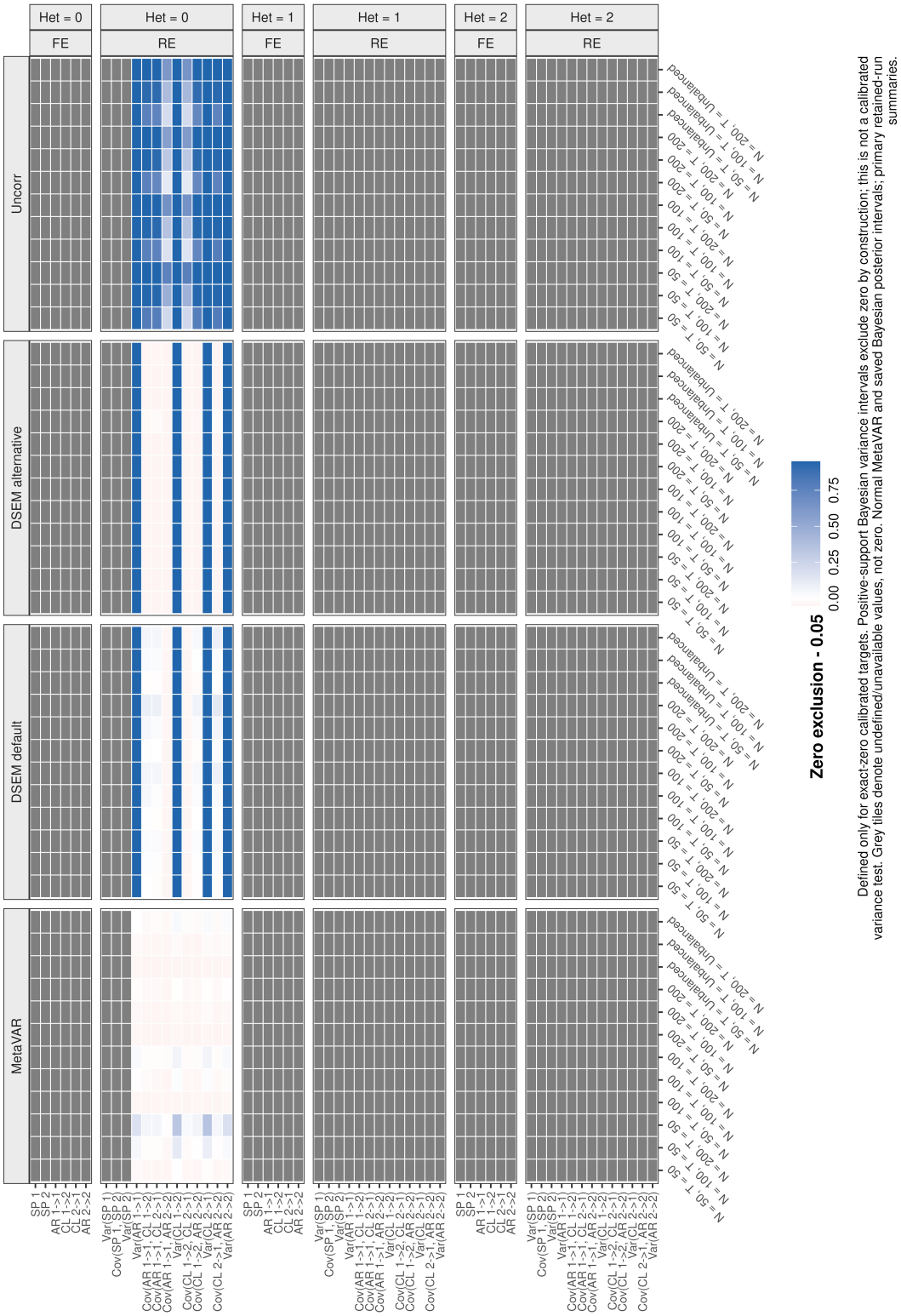


Figure 6
Interval Exclusion for Exact-Zero Targets (Bayesian Variance Intervals Exclude Zero by Construction)



Defined only for exact-zero calibrated targets. Positive-support Bayesian variance intervals exclude zero by construction; this is not a calibrated variance test. Grey tiles denote undefined/unavailable values, not zero. Normal MetaVAR and saved Bayesian posterior intervals; primary retained-run summaries.